

## Session 4: PAKE-0, Assumptions, and Macros

Slides prepared by UMBC Protocol Analysis Lab

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# Passwords

**Password:**

Log In

[Forgot Your Password?](#)

Can help prove your identity to somebody.

We've used these forever.

Should be kept secret and hard to guess.

Often memorized. . .

Don't store these in plain-text anywhere.

# PAKE-0

Alice

Bob

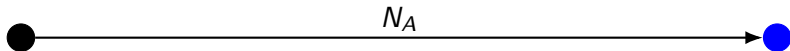
shared secret password: **pw**

● = Reception    ● = Transmission

# PAKE-0

Alice

Bob



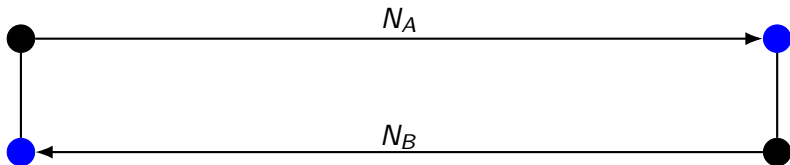
shared secret password: **pw**

● = Reception      ● = Transmission

# PAKE-0

Alice

Bob



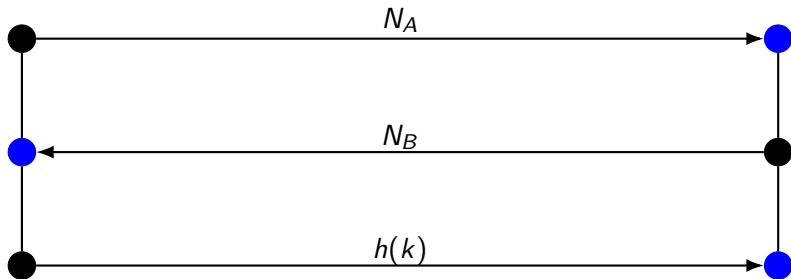
shared secret password: **pw**

● = Reception      ● = Transmission

# PAKE-0

Alice

Bob



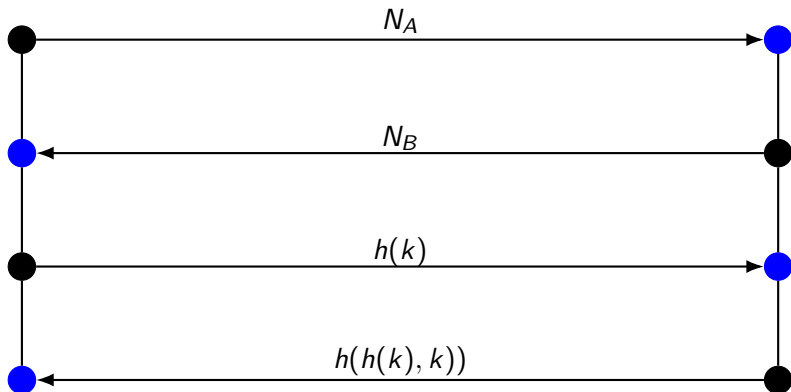
shared secret password: **pw**  $k = h(pw, Alice, Bob, N_A, N_B)$

● = Reception    ● = Transmission

# PAKE-0

Alice

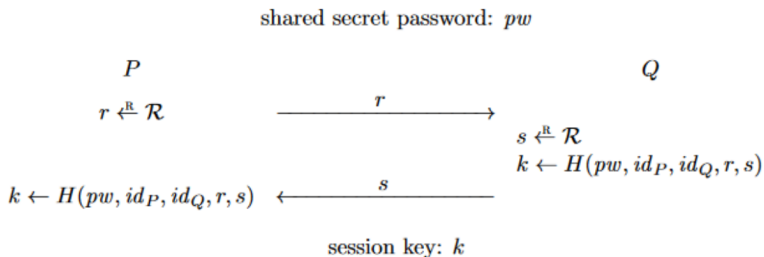
Bob



shared secret password: **pw**  $k = h(pw, Alice, Bob, N_A, N_B)$

● = Reception    ● = Transmission

## PAKE-0 textbook diagram

Figure 21.13: Protocol  $\text{PAKE}_0$ 

**Figure:** Boneh, Dan, and Victor Shoup. “A graduate course in applied cryptography.” Draft 0.5 (2020).

# Assumptions in CPSA (1)

Data origination assumptions:

**uniq-orig** (e.g., nonce)

## Definition (unique origination)

A value *uniquely originates* in a bundle if it originates on exactly one strand. Note that if the unique origination point of a value is on a regular strand, then it cannot be produced in a “create” instance because that would constitute a second point of origination.

# Assumptions in CPSA (2)

**non-orig** (e.g., private key)

## Definition (non-origination)

A value is *non-originating* in a bundle if it does not originate on any strands. A simple lemma can show that a value is non-originating only if it is not carried in any message in the bundle.

## Assumptions in CPSA (3)

**pen-non-orig** (e.g., pre-shared password)

### Definition (penetrator non-originating)

A value is *penetrator non-originating* in a bundle of a protocol if it does not originate on any penetrator strands. Clearly, such a value cannot be produced in a “create” instance.

# In-Class Activity: Model PAKE-0 in Small Groups

Now that you've seen the model, we will break you into groups with a TA assigned to each group to help and check your work. You will model this protocol using our new knowledge assumptions. instance.

# Homework

# Model more in CPSA

# Model more in CPSA

Now that you've seen PAKE modeled, can you model a more challenging protocol?

Model your meeting protocol from Homework 1!

# Survey

Due at 6:00pm on Tuesday the 28th

